FYBA ECONOMICS SEM 2

CIA -2 ASSIGNMENT

UAECO4502CR1

ECONOMICS PROJECT

BANDAR MIFFLIN Poster Company

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INDEX

- Team Members…… …….03
- Business Outline…….……..04
- Production process……...…….……05
- Market Survey Analysis…….…..…….06
- Price Discrimination…….…..……....08
- Elasticity Of Demand…….……...……10
- Break Even Point……...……12
- Cost Structure…….……15
- Cost Minimization…….…….……16
- Sales Maximisation……....……19
- Profit Maximisation…….……..23
- Other Relevant Factors……...…..…….…….26
- Limitations Of The Business…….……28
- Limitations Of Analysis…….…28
- Key Performance Indicators……..29
- Conclusion…….....29
- Acknowledgement……...……..29
- Appendix……...30

Team Members

1. Basil Nainan Shibu
   Chief Financial Officer
   - ➢ Responsible for all accounting, maintaining accounts of the business
   - ➢ Responsible for calculating cost minimisation.

2. Chris Kurian Joseph
   Chief Sales Officer
   - ➢ Responsible for discovering new markets and exploring new sales opportunities.
   - ➢ Responsible for conducting price discrimination.

3. Jonathan Flettin Marattukalam
   Founder, Chief Marketing Officer
   - ➢ Responsible for the management of all sorts of promotional media and changing tastes of the customers.
   - ➢ Responsible for calculating sales maximisation, assisting with accounting, and producing financial reports

4. Sidharth S
   Chief Executive Officer
   - ➢ Responsible for supervision and all technical aspects relating to upscaling and the final prints.
   - ➢ Responsible for calculating profit maximisation and Break Even point.

Business Outline

Bandar Mifflin is a student-run business catering to the college community's desire for visually appealing memorabilia from their favourite TV shows, movies, video games and music artists. We provide the sale of high quality A3 sized posters at an affordable price. Our posters undercut the market rate by more than 40%[1] while retaining a 65% profit margin on each unit of sale. Bandar mifflin is easily scalable, with minimal fixed costs, our business can branch out into different colleges and their respective fests. We have fostered business relations with suppliers around the Fort area, and have come to agreements with regards to the price of printing and packaging which stands as our main variable cost.

In terms of scalability, rather than just focusing on direct offline sales, we expanded to other markets. We sold throughout Mumbai more specifically Grant Road, Powai (IIT Bombay), and Cuffe Parade, while also expanding into Bangalore, selling in Christ (deemed to be) University.

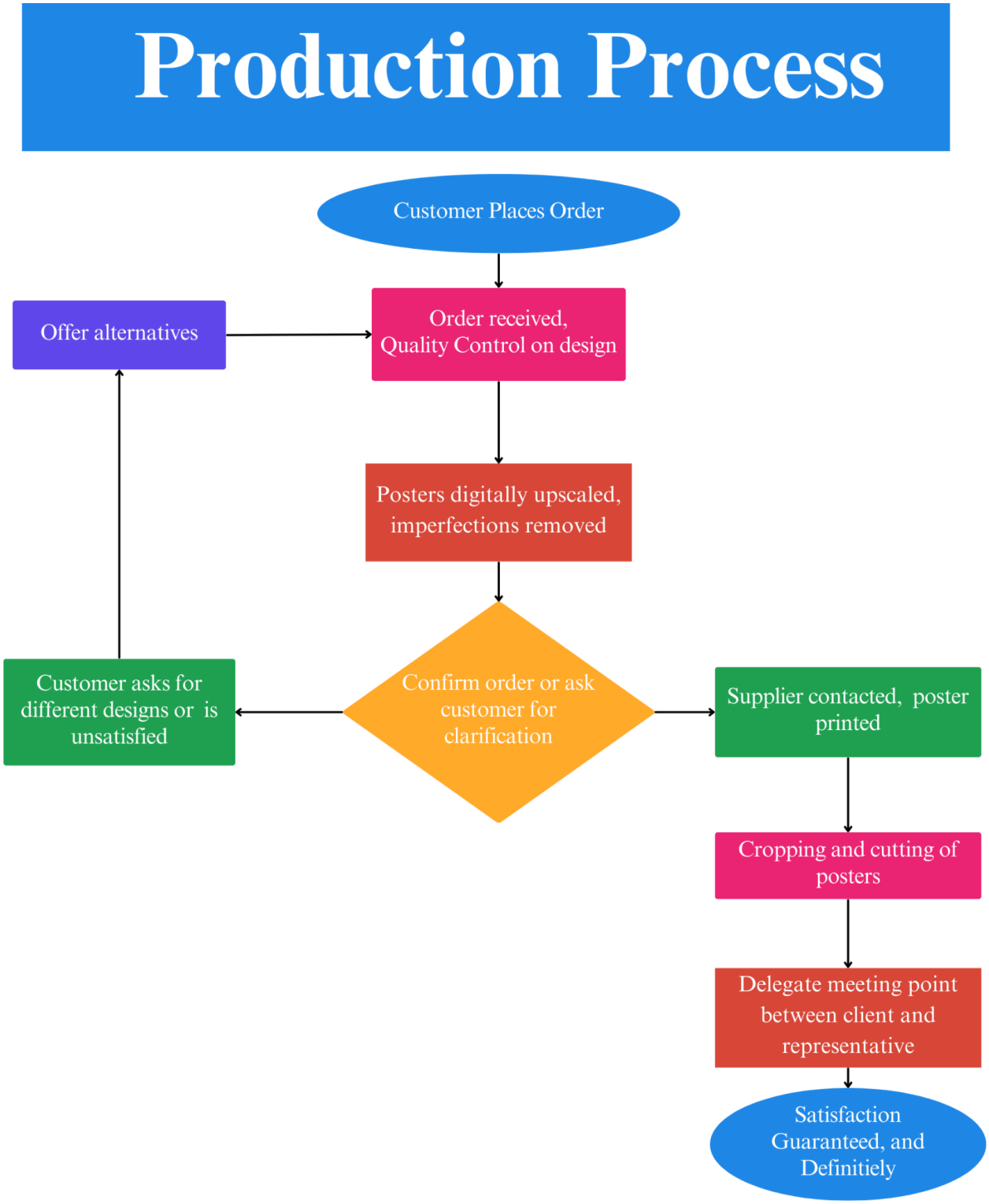
In terms of Social Media, both members of sales and marketing were incredibly active. We focused on 3 posts a day showcasing the various designs we would be able to print. We procured these designs from copyright-free copyright-free sites such as The Movie Database, pinterest, as well as making our designs. We decided what we would post and what topic to cover based on the market survey results. As a result of our social media endeavours, we amassed 74 followers, and were able to reach 253 accounts in total. Additionally, 65 users engaged with our content, and we garnered a total of 3066 impressions.[2]

Operating with this business model in mind, Bandar Mifflin has successfully sold 146 units, reaching a total revenue of ₹16,070.

We operate on the motto *"Satisfaction guaranteed, and definitely"*

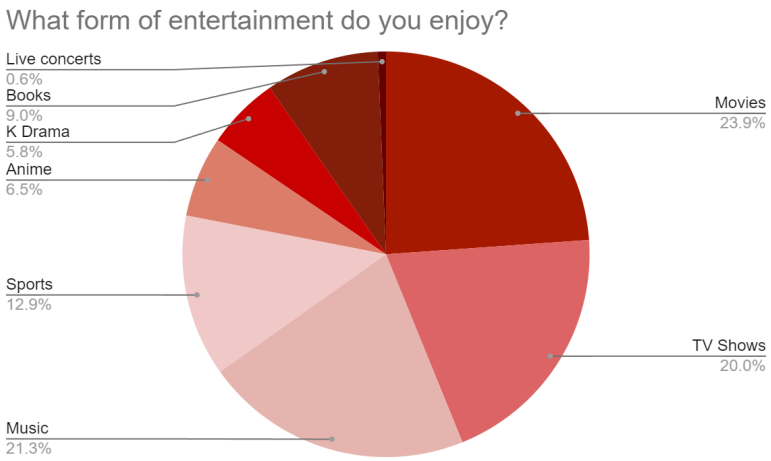
[1] *Market rate: ₹200 (approx.) according to various online websites*
[2] *Appendix B*

The following is a run-down of our production process:

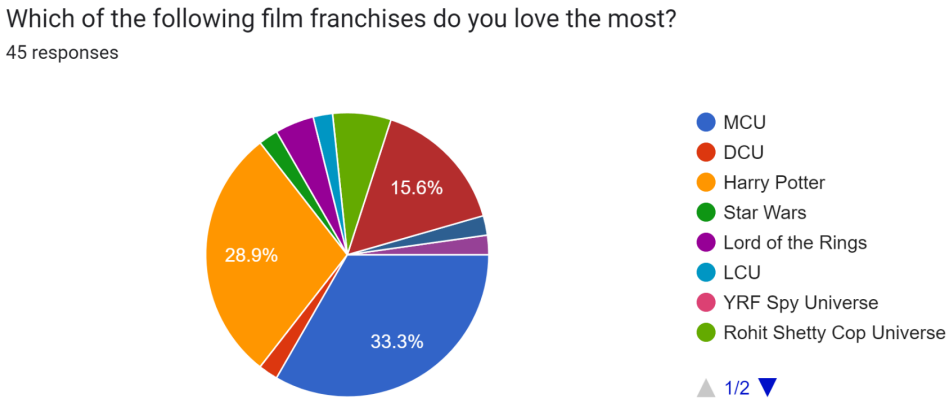


Market Survey Analysis

This survey took place in December to understand what sort of designs our customers would prefer and at what price they were most likely to buy the posters. We were able to understand that the main entertainment industries we have to focus on are movies, TV shows and music.



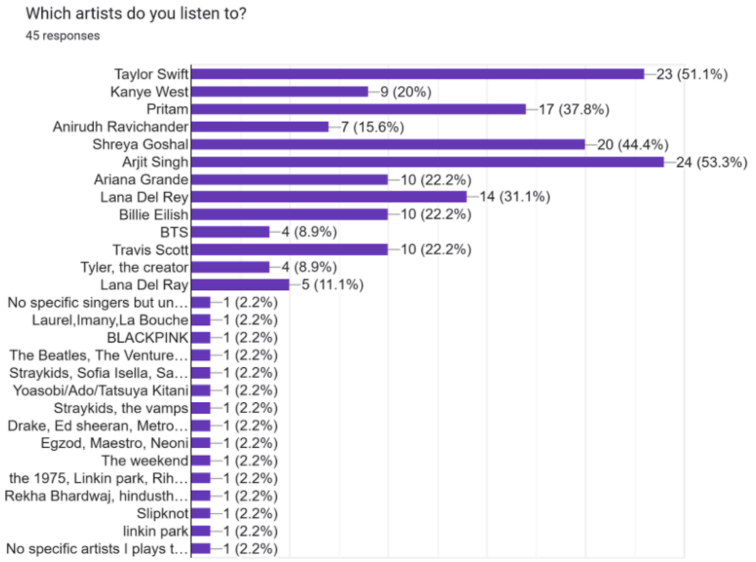
Since a large number of respondents primarily watched movies, it would be of use to know which franchise they loved the most. In the current landscape of Indian and international cinema, all studios have launched their own film franchises and universes which poses a challenge for us in handpicking designs.



The results show that the Marvel Cinematic Universe (MCU) retains the maximum number of fans with a 33.3% of the share of respondents.

The second most popular franchise on the list was the Harry Potter series. Even though the movies got over a long time ago, the fans of the series are still active. To print posters for these franchises would be a difficult task as these movies aren't really known for their posters and we would have to rely on fan art.

The next industry to tackle was the music industry. As expected, the leading artist was Taylor Swift, with 51.1% of survey respondents choosing her as one of their favourite artists.



With almost everything covered, we just had one final question to be answered, what would be the price range that people would be most comfortable with? The answer was obvious, people would go for the lower range of 120-150 rupees and this holds true to the law of demand. However there were a few decisions to be made by the firm, there were three types of paper and two sizes for us to possibly sell our posters, which included 180 GSM, 210 GSM and 300 GSM, as well as A3 and A4 sizes and so we modified our question.

We added that the difference in price was due to the difference in size and thickness, a higher price would get thicker paper and a bigger size. This led to some interesting results with 37.8% of the people, which is about 17 respondents, deciding to pay 120-150 rupees while a higher number of 21 respondents which was 46.7%, decided to pay a higher rate of 150-180. Not only did this allow us to finalise on the A3 size and 210 GSM paper for now, it also allowed us to set our price as 150 rupees.

Overall, this survey was able to analyse the current market trends, we were able to find out that people were still nostalgic for old shows or movies. By selling our posters, we potentially help them relive their favourite movie moments.

Price Discrimination

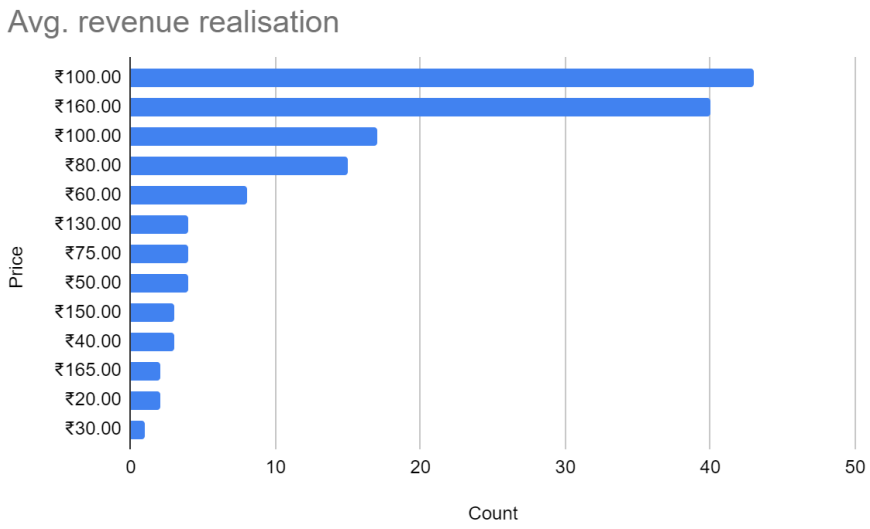
Price discrimination is the method practised by a firm, where it charges different prices for the same product based on various factors such as market elasticity, the quantity at which the product is being bought and more. There are three types of price discrimination, namely 1st degree, 2nd degree and finally 3rd degree price discrimination.

Total Number of Customers : 100+

Markets we operated in:

1. St.Xavier's College
2. IIT Bombay
3. Grant Road
4. Christ University (Bangalore)
5. Social Media

Breakdown of Average revenue by count



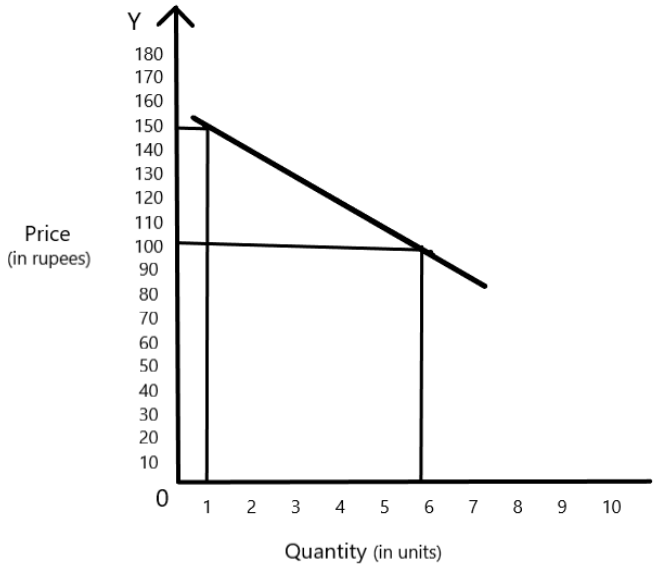
First Degree Price Discrimination is the type where we charge the highest amount that a customer is willing to pay. This usually involves the firm knowing that particular customer in detail, with regards to their financial background and other factors. In our case, most of our customers were either people that were our acquaintances and/or friends. Due to this, in most of the cases we were aware that our customers wouldn't be willing to buy at our initial rate of Rs.150, due to budget constraints or other reasons, and so we changed the prices according to a range of Rs.120-140.

Another factor which helped us understand what price we could charge for different customers was a customer survey. It showed us that people were willing to pay anything from Rs.120 all the way to Rs.180. This gave us an insight into how much we could stretch out our prices depending on the customer.

Some of the examples for first degree price degree discrimination are as follows. The first is in the case of Customer A. Since he was a friend we got to know that the maximum he was willing to pay was Rs.130, and since he lived afar he also paid a Rs.40 delivery fee and so we can see how both first and third degree price discrimination(different market so higher price) has taken place. Other cases where we charged a similar or same price were in the case of Customer B(130), Customer C(130), Customer D (120), and Customer E (100).

By exercising such price discrimination, we were still able to make a revenue while at the same time it did not affect our profit margins as much. Furthermore, due to the flexibility of our prices, a few of these customers (almost two to five) were interested in buying a higher quantity and at the same time even returned to the firm as second time customers.

Second Degree Price Discrimination is the type of price discrimination where price has been altered according to the quantity that has been bought. This was exercised in our stall in the form of a Buy 2 Get 1 offer, Customer F was able to avail this offer, he bought six posters, of which he paid for four, and got two for free. His total bill amounted to Rs.600. However, this method of price discrimination was not too effective for the firm to get sales and so we resorted to cutting down the price to Rs.100. Another situation where the firm exercised 2nd degree price discrimination was when we charged Rs.140 for a customer in Bangalore instead of our initial rate of Rs.Rs.160 as he was willing to buy 5 posters at once.



Third Degree Price Discrimination is where different rates are charged according to the elasticity of demand. Our initial rate and market was Rs.150 at our online platform.
Our second rate and market was at our stall in St. Xaviers college, where at first we charged 100 rupees and then on the second day changed it to Rs.60 and even Rs.40. Our final and third market was Christ University in Bangalore, where we were able to charge around Rs.160, this price was mostly charged due to delivery charges and the commission to be paid to our agent in Bangalore.

Each of these markets have had a different effect on our business. With the help of the 150-160 price range we were able to ensure maximum revenue or profits for our firm. We were able to make around Rs.4800 from the Bangalore order which contributes to around 33% of the total revenue made by the firm by selling 40 posters. However it was at the stall that the firm was able to achieve sales maximisation. We made around the same amount of money but were able to sell our remaining inventory, 77 posters in total.

Elasticity Of Demand

Price elasticity of a product is a measurement of the change in demand for a good or service in relation to a change in its price. In our case, in order to measure price elasticity we will be taking the example of sales made by Bandar Mifflin when we set up a stall in the woods, where we sold at two different rates, Rs.100 and Rs.80.

$$\text{Price Elasticity Formula} = \frac{Percentage\ change\ in\ Quantity\ Demanded}{Percentage\ change\ in\ Price}$$

Price	**Quantity Demanded**
Original Price = 100	43
New Price = 80	80

$$e_{dp} = \frac{\%\Delta Q}{\%\Delta P}$$

$$= \frac{\Delta Q}{Q} \times \frac{P}{\Delta P}$$

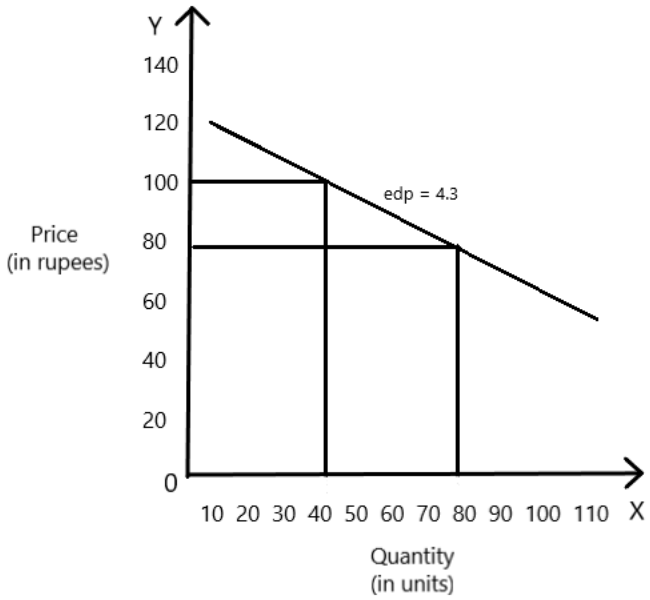
$$= \frac{(80-43)}{43} \text{ x } \frac{100}{(80-100)}$$

$$= \frac{37}{43} \text{ x } \frac{100}{20}$$

$$= 0.86 \text{ x } -5$$

$$= 4.3$$

$e_{dp} = 4.3$, Hence our product is very elastic, with a small change in price, we were able to double the amount of posters sold.



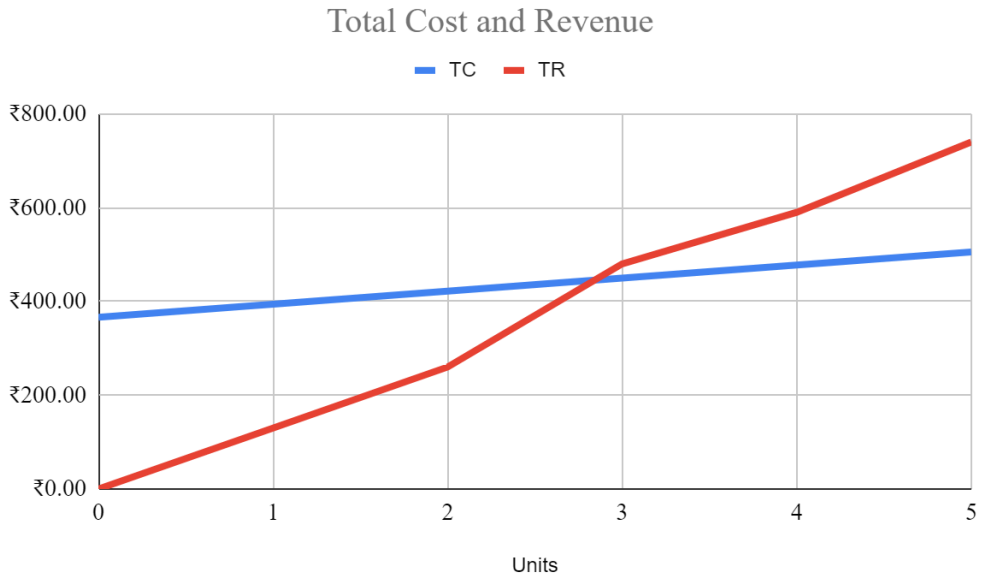
Break Even Point

The first concern for any budding business is to attain break even, i.e to recover the investment in fixed capital.

B.E.$P_Q = \frac{F}{P - V}$ where,

F = Fixed Cost,
P = Price (Average Revenue)
V =Average Variable Cost

Based on our sales data, we broke even on the third unit, by achieving the first positive profit of Rs. 30.65. The given chart shows actual revenue and cost information to point out Break Even.



Since there is very low variance on our variable cost and the business doesn't demand high capital investment, we have projected multiple break-even points by fluctuating Selling price at Rs. 80, Rs.100, Rs.120 and Rs.150.

Fixed Cost is taken as Rs.366, and Variable Cost (Weighted Average) as Rs.22.349 ~ Rs. 22.3

BEP @ Price = Rs.80

$$BEP_Q = \frac{F}{P-V}$$

$$= \frac{366}{80-22.3}$$

$$= \frac{366}{57.7} = 6.34\ units$$

BEP @ Price = Rs.100

$$BEP_Q = \frac{F}{P-V}$$

$$= \frac{366}{100-22.3}$$

$$= \frac{366}{77.7} = 4.71\ units$$

BEP @ Price = Rs.120

$$BEP_Q = \frac{F}{P-V}$$

$$= \frac{366}{120-22.3}$$

$$= \frac{366}{57.7} = 3.75\ units$$

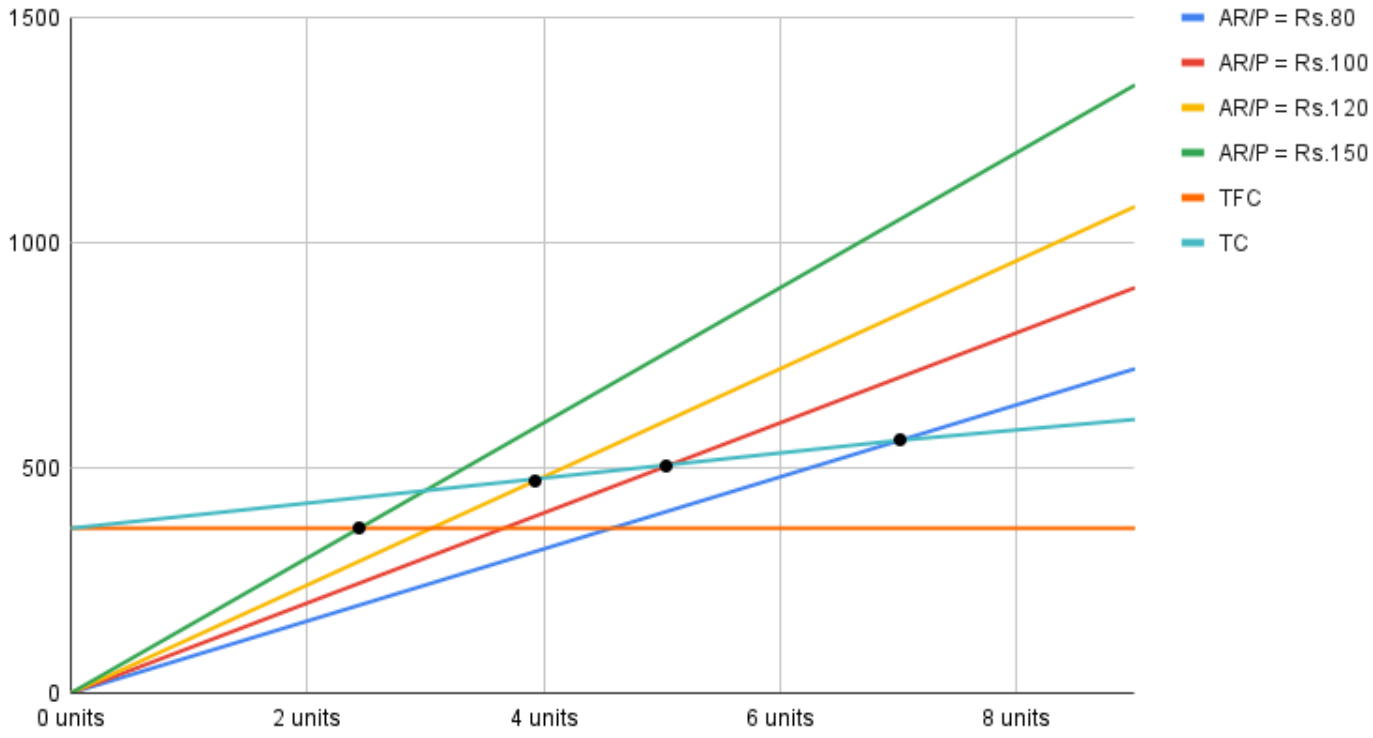
BEP @ Price = Rs.150

$$BEP_Q = \frac{F}{P-V}$$

$$= \frac{366}{150-22.3}$$

$$= \frac{366}{57.7} = 2.86\ units$$

Visual Plotting of Break-Even Point at 4 different prices



Cost Structure

Fixed Costs

The fixed costs in our business include :

1. *Research and Development*: We undertook a proper R&D process in the development of our product by testing samples, getting user opinion, and testing other factors like printing size, quality, etc.

2. *Stationery*: Pens, notebooks, cutters, packaging tape, UPI QR codes, name tags for each member and pamphlets of our business were purchased

3. *Marketing and Promotion*: We gave away four free posters and some chocolates for marketing and promotion.

Variable Costs

➢ The variable cost can be broken down into 3 segments, printing, cutting and packaging.

➢ Among these, printing is the biggest expense at an average of Rs.20, we tried different qualities of paper and different service providers to find the best cost.

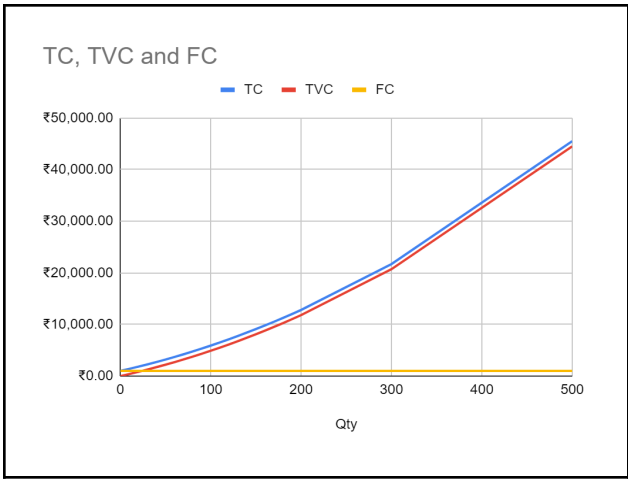
➢ Cutting implies the cutting of posters to desired size, and removing imperfections. For packaging, we used transparent plastic covers. These particulars make up the rest of the variable cost. The Cost of both cutting and packaging decreased as the business progressed.

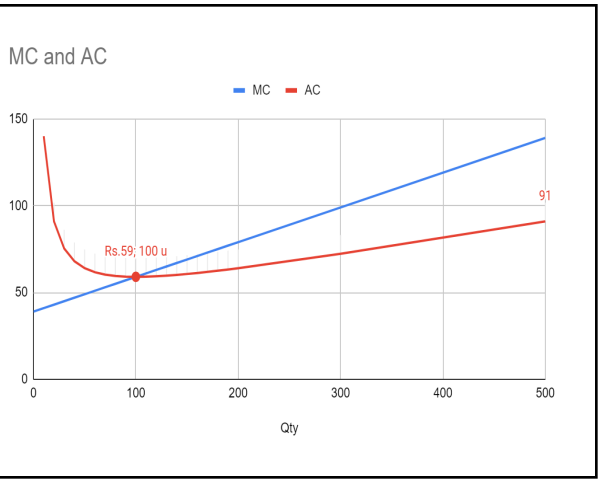
➢ We reaped the benefits of economies of scale by increasing our level of output by printing in bulk. This can be seen by the decreasing values of marginal cost.[3] Marginal Cost fell from Rs.28 to Rs.22 as output increased.

[3] *Appendix A*

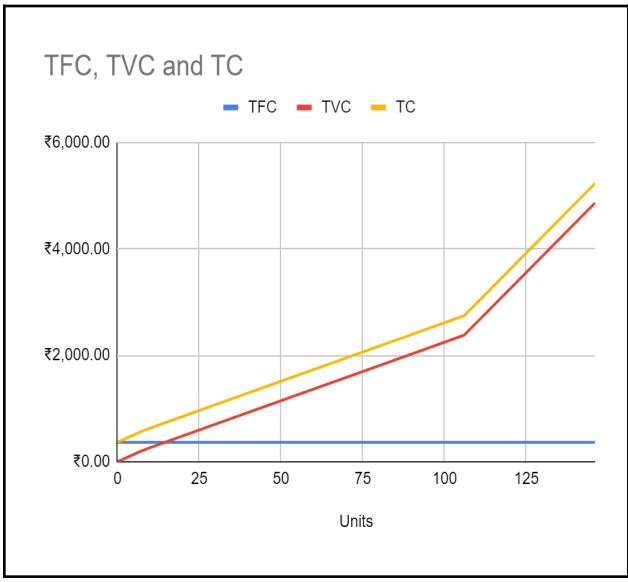
COST MINIMISATION

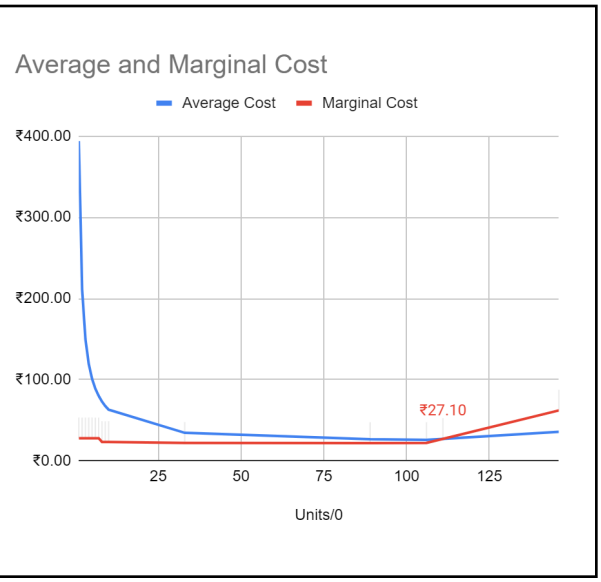
Projected Cost Curves





Actual Cost Curves[4]





[4] Appendix A



Cost Schedule

Units	TFC	TVC	TC	AC	MC
0	₹366.00	₹0.00	₹366.00	-	-
1	₹366.00	₹27.85	₹393.85	₹393.85	₹27.85
2	₹366.00	₹55.70	₹421.70	₹210.85	₹27.85
3	₹366.00	₹83.55	₹449.55	₹149.85	₹27.85
4	₹366.00	₹111.40	₹477.40	₹119.35	₹27.85
5	₹366.00	₹139.25	₹505.25	₹101.05	₹27.85
6	₹366.00	₹167.10	₹533.10	₹88.85	₹27.85
7	₹366.00	₹194.95	₹560.95	₹80.14	₹27.85
8	₹366.00	₹218.28	₹584.28	₹73.04	₹23.33
9	₹366.00	₹241.61	₹607.61	₹67.51	₹23.33
10	₹366.00	₹264.94	₹630.94	₹63.09	₹23.33
20	₹366.00	₹486.45	₹852.45	₹42.62	₹22.15
30	₹366.00	₹709.23	₹1,075.23	₹35.84	₹22.28
33	₹366.00	₹774.16	₹1,140.16	₹34.55	₹21.64
40	₹366.00	₹927.16	₹1,293.16	₹32.33	₹21.86
50	₹366.00	₹1,147.31	₹1,513.31	₹30.27	₹22.02
60	₹366.00	₹1,361.80	₹1,727.80	₹28.80	₹21.45
70	₹366.00	₹1,582.10	₹1,948.10	₹27.83	₹22.03
80	₹366.00	₹1,802.41	₹2,168.41	₹27.11	₹22.03
89	₹366.00	₹2,002.90	₹2,368.90	₹26.62	₹22.28
90	₹366.00	₹2,025.60	₹2,391.60	₹26.57	₹22.70
100	₹366.00	₹2,249.20	₹2,615.20	₹26.15	₹22.36
106	₹366.00	₹2,378.60	₹2,744.60	₹25.89	₹21.57
110	₹366.00	₹2,654.09	₹3,020.09	₹27.46	₹68.87
120	₹366.00	₹3,247.28	₹3,613.28	₹30.11	₹59.32
130	₹366.00	₹3,868.71	₹4,234.71	₹32.57	₹62.14
140	₹366.00	₹4,493.17	₹4,859.17	₹34.71	₹62.45
146	₹366.00	₹4,863.00	₹5,229.00	₹35.82	₹61.64

Estimation of Cost Equation

Cost minimisation is a financial strategy used by firms to determine the most cost-effective way of providing goods and services while maintaining a desired level of quality. It involves finding the perfect combination of labour and capital that produces output at the lowest cost.

Here, we have estimated the cost functions by finding a line of best-fit based on our educated estimates to determine the projected cost minimisation conditions.

Assumptions:

Total Fixed Cost(TFC) = Rs. 1000

TC equation, $Y = ax^2 + bx + c$; $TVC = ax^2 + bx$

where, C = Fixed Cost

Y = Total Cost

X = Quantity

After conducting a survey to outsource printing and other costs, we came up with the following equations:

1) When Q = 10 ; TVC = 400 $\rightarrow$ $a(10)^2 + b(10) = 400$ $\rightarrow$ $100a+10b=400$

2)When Q = 100 ; TVC = 5000 $\rightarrow$ $a(100)^2 + b(100) = 5000$ $\rightarrow$ $10000a+100b=5000$

After solving for a and b, we find that a = 0.1 ; b = 3.9

Hence, $TVC = 0.1q^2 + 39q$

$TC = 1000 + 0.1q^2 + 39q$

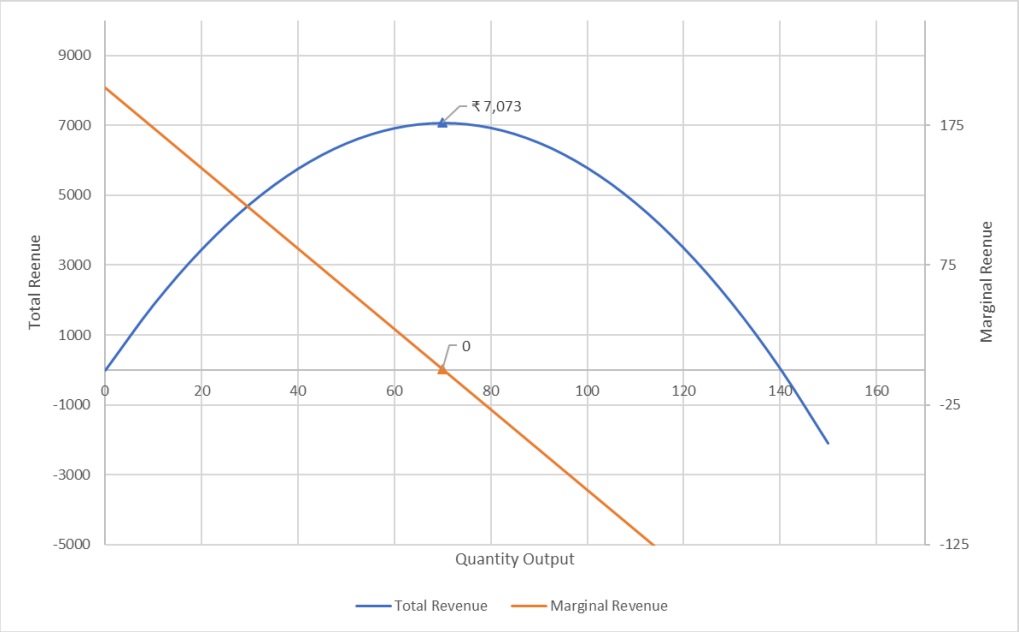
$MC = 0.2q + 39$

Comparison of projections vs. reality

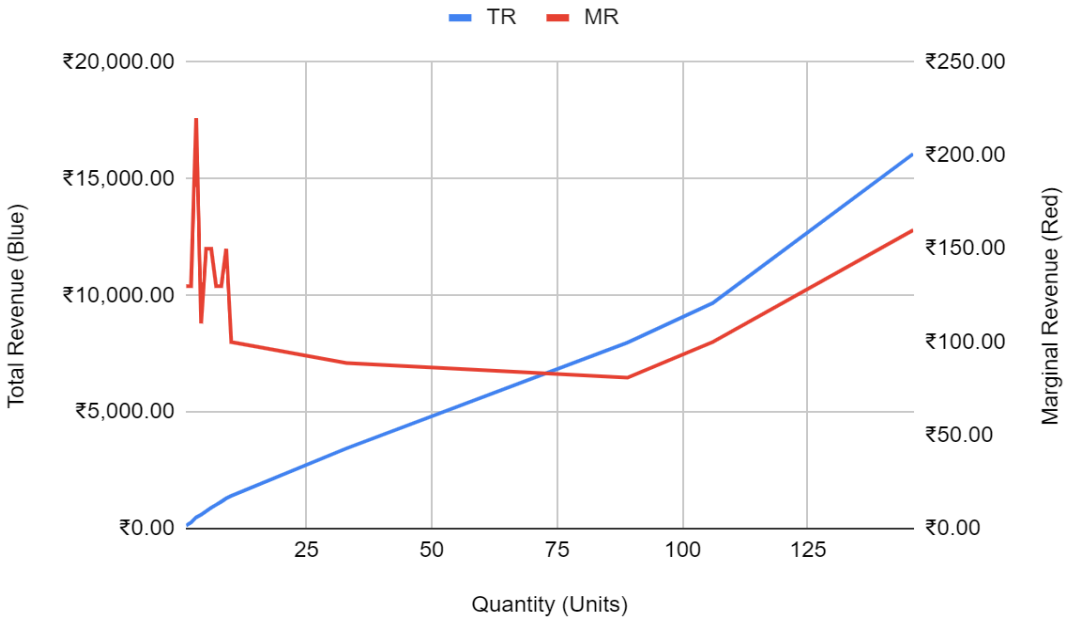
- Our estimated cost was very high. In reality, we were able to produce at much lower cost. This is applicable to both fixed and variable cost.
- Our actual fixed cost was only Rs.366 unlike Rs.1000 we estimated primarily because we had planned an offline stall for a longer term that demanded fixed capital. Variable Costs too fell as sales soared, printing costs fell and many other planned production-related expenses were never incurred.
- Based on actual data, we were able to determine a cost-minimisation point as the MC and AC curves intersect at the 111th unit. Marginal Cost rose after the 106th unit and units afterwards included commission payments.

SALES MAXIMISATION

Projected Revenue Curve



Actual Revenue Curve



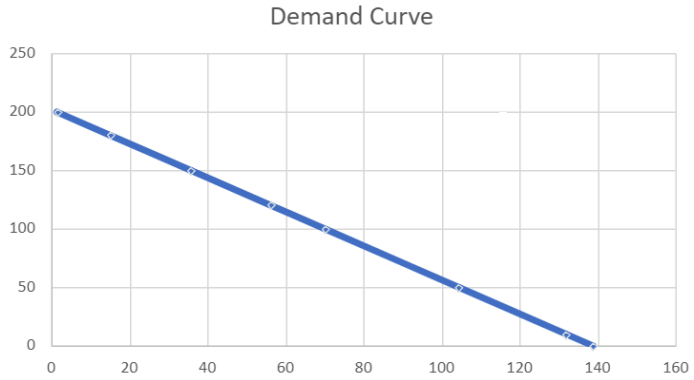
Revenue Schedule

Units	TR	AR	MR
0	₹0.00	-	-
1	₹130.00	₹130.00	₹130.00
2	₹260.00	₹130.00	₹130.00
3	₹480.00	₹160.00	₹220.00
4	₹590.00	₹147.50	₹110.00
5	₹740.00	₹148.00	₹150.00
6	₹890.00	₹148.33	₹150.00
7	₹1,020.00	₹145.71	₹130.00
8	₹1,150.00	₹143.75	₹130.00
9	₹1,300.00	₹144.44	₹150.00
10	₹1,400.00	₹140.00	₹100.00
20	₹2,297.43	₹114.87	₹89.74
30	₹3,174.53	₹105.82	₹87.71
33	₹3,440.00	₹104.24	₹88.49
40	₹4,020.15	₹100.50	₹82.88
50	₹4,833.93	₹96.68	₹81.38
60	₹5,640.70	₹94.01	₹80.68
70	₹6,437.50	₹91.96	₹79.68
80	₹7,264.31	₹90.80	₹82.68
89	₹7,970.00	₹89.55	₹78.41
90	₹8,073.67	₹89.71	₹103.67
100	₹9,068.97	₹90.69	₹99.53
106	₹9,670.00	₹91.23	₹100.17
110	₹10,330.02	₹93.91	₹165.01
120	₹11,911.58	₹99.26	₹158.16
130	₹13,523.00	₹104.02	₹161.14
140	₹15,133.70	₹108.10	₹161.07
146	₹16,070.00	₹110.07	₹156.05

Projected Sales Maximisation

Sales maximisation can be defined as the quantity a firm is producing at which the change in total revenue with respect to a change in output is zero ($\frac{dTR}{dQ} = MR = 0$). Effectively, at this point, the firm is producing where it is maximising revenue and therefore, sales.

To find this point, one must first derive a demand curve from a demand schedule. Taking into consideration the results from the market survey, we performed regression analysis on the X and Y axes (Quantity and Price respectively), to obtain a regression equation for the demand curve.



$$P = 201.6651 - 1.4374Q$$

(To derive the Total Revenue)(TR) curve, we use the relation $TR = P{\cdot}Q$

$P = 201.6651 - 1.4374Q \Rightarrow TR = 201.6651Q - 1.437Q^2$

Marginal Revenue, $MR = \frac{dTR}{dQ} = 201.6651 - 1.437Q$

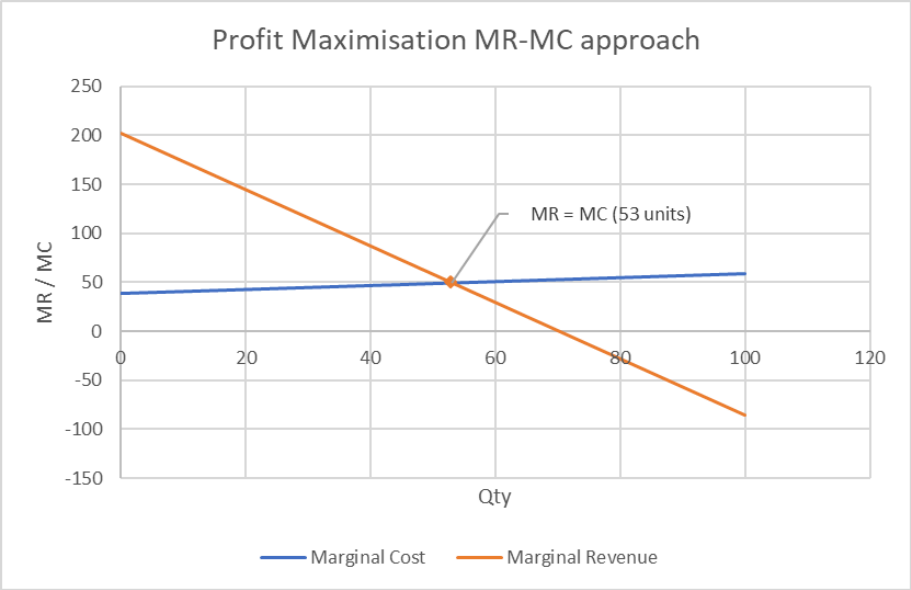
The condition for sales maximisation as stated earlier, was that $MR = 0$. Therefore, if we substitute MR for 0, we get the sales maximising value of output which is 70.1493≈**70 units**. At which **₹7,073** is made in Total Revenue.

Actual Sales Maximisation

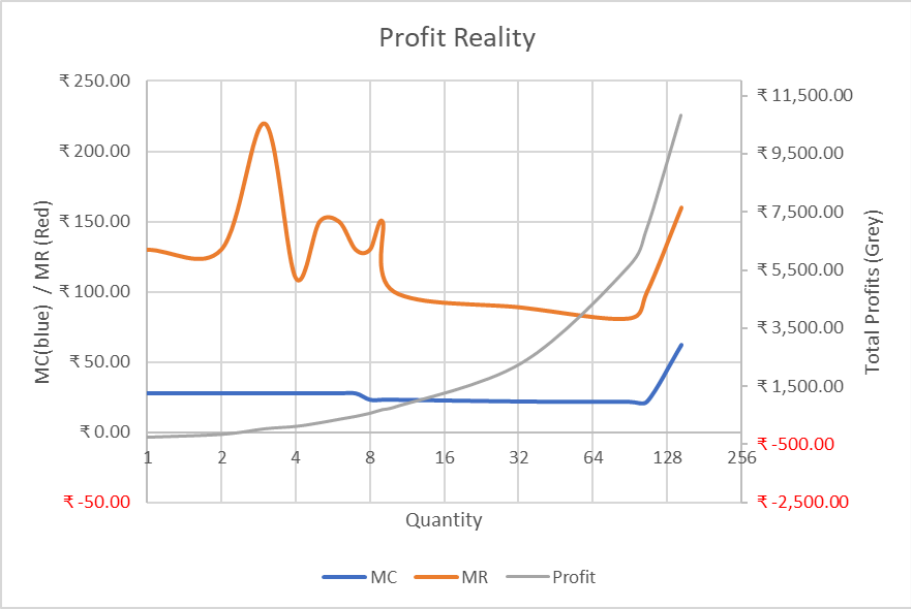
- With the given data and graph, it can be observed that the sales maximisation point cannot be determined as MR ≠ 0.
- We far surpassed our projected sales maximising quantity of 70 units, producing more than double at 146 units. Additionally, a total revenue of ₹16,070 was generated which is also more than double of our projected total revenue at the sales maximisation point of ₹7,073.
- The prime reason for the low projections are that the estimates were made with the college as the only market. However, we were able to expand to other markets due to which we were able to sell more quantity.
- It can be observed that our marginal revenue curve (actuals) is very erratic in the beginning. This is because we were experimenting with prices and most sales were of small quantities with price discrimination. Later on, our prices stabilised, leading to a smoother Marginal Revenue curve.
- As we stopped business at 146 units, our Total Revenue curve was still rising at a strong positive slope, suggesting that more sales are possible.

PROFIT MAXIMISATION

Projected Profit Max



Actual Profit Max



Profit Schedule

Units	Profit	Marginal Profit	MC	MR
0	-₹366.00	-	-	-
1	-₹263.85	₹102.15	₹27.85	₹130.00
2	-₹161.70	₹102.15	₹27.85	₹130.00
3	₹30.45	₹192.15	₹27.85	₹220.00
4	₹112.60	₹82.15	₹27.85	₹110.00
5	₹234.75	₹122.15	₹27.85	₹150.00
6	₹356.90	₹122.15	₹27.85	₹150.00
7	₹459.05	₹102.15	₹27.85	₹130.00
8	₹565.72	₹106.67	₹23.33	₹130.00
9	₹692.39	₹126.67	₹23.33	₹150.00
10	₹769.06	₹76.67	₹23.33	₹100.00
20	₹1,444.98	₹67.59	₹22.15	₹89.74
30	₹2,099.30	₹65.43	₹22.28	₹87.71
33	₹2,299.84	₹66.85	₹21.64	₹88.49
40	₹2,726.99	₹61.02	₹21.86	₹82.88
50	₹3,320.62	₹59.36	₹22.02	₹81.38
60	₹3,912.90	₹59.23	₹21.45	₹80.68
70	₹4,489.40	₹57.65	₹22.03	₹79.68
80	₹5,095.90	₹60.65	₹22.03	₹82.68
89	₹5,601.10	₹56.13	₹22.28	₹78.41
90	₹5,682.07	₹80.97	₹22.70	₹103.67
100	₹6,453.77	₹77.17	₹22.36	₹99.53
106	₹6,925.40	₹78.61	₹21.57	₹100.17
110	₹7,309.93	₹96.13	₹68.87	₹165.01
120	₹8,298.30	₹98.84	₹59.32	₹158.16
130	₹9,288.29	₹99.00	₹62.14	₹161.14
140	₹10,274.53	₹98.62	₹62.45	₹161.07
146	₹10,841.00	₹94.41	₹61.64	₹156.05

Derivation of Projected Profit Maximisation

The primary objective of any firm in a competitive market is profit maximisation. Since we have already derived the TR function and the TC function, we shall approach profit maximisation with the TR-TC approach.

$TR = 201.6651Q - 1.437Q^2$

$TC = 0.1Q^2 + 39Q + 1000$

$Total\ Profit,\ \pi = TR - TC$

$\pi = 201.6651Q - 1.4374Q^2 - [0.1Q^2 + 39Q + 1000]$

$\pi = -1.5374Q^2 + 162.6651Q - 1000$

Since the profit function is quadratic, at the point of profit maximisation, marginal profit will be zero. ($M\pi = 0$). Taking the first derivative of the profit function and equating it to zero shall thus give us the quantity at which profit can be maximised.

$M\pi = \frac{d\pi}{dQ} = 162.6651 - 3.0748Q$

Taking Marginal Profit as zero,

$0 = 162.6651 - 3.0748Q$

$Q = 52.9 \sim 53$

Evidently, the firm will attain profit maximisation when it sells 53 units, where total profit would be ₹3302.69, giving a per unit profit of ₹62.3

Actual Profit Maximisation

- ➢ In reality, our MR and MC curves never intersect, and profits were rising exponentially.
- ➢ We can observe that our MC curve is rather flat owing to minimal fluctuations on our cost structure, but MR is erratic as we experimented with different prices and markets. Throughout our business, Marginal Revenue was well above Marginal Cost.
- ➢ The total profit curve indicated by the grey line on the graph stands as proof of our consistent growth in profit. Our business has been able to maintain a gross profit margin of 65%.
- ➢ In reality, we were able to beat the expected point of profit maximisation of 53 units, where we had rising profits at almost triple the quantity.

Other Relevant Factors

❖ **Value Added Services - Delivery**

➢ The concept of delivery was an unexpected addition to the revenue. As we scaled the business, we realised that the external (beyond SXC) market had high demand. So, we, on a trial basis, started delivery of our products in a short-radius around the college. A nominal fee was charged for the same. Later, we expanded beyond Mumbai by shipping to Bangalore in bulk.

➢ We realised that delivery is not feasible for small, retail orders, as it brings the price point too high. However, it was feasible for us to waive delivery fees for bulk orders.

➢ We delivered 9 posters around Mumbai as retail orders at an average delivery fee of Rs.20. For our bulk shipment of 40 posters to Bangalore, we waived delivery fees, without compromising on our margins, taking advantage of economies of scale.

❖ **Value Added Services - Customization**

➢ The primary USP (Unique Selling Proposition) of our business is customization. No mainstream poster seller offers customization capabilities.

➢ The customer can send designs they like and request minor modifications both in design and printing.

❖ **Commission Sales**

➢ With only 4 employees, a sales bottleneck was bound to happen which is why we initiated a commission-based sale system where one could sign up as an agent, to procure orders which we will process, print and handover to said agent, who shall take care of delivery and payment handling.

➢ We had our first agent in Bangalore, who made 1 bulk order of 40 posters. The selling price of the posters were Rs.160, of which Rs.40 flat rate commission was offered.

➢ Such a system enables us to expand our servicing locations at low capital investment as we have high profit margins to play with.

❖ **Offline vs. Online**

- ➢ Our business operated both offline (as a stall in the woods) and online (as an instagram page).
- ➢ Online orders were few, but we were able to sell at a higher price and offer greater customization. However, organic growth online was hard and demanded greater investment of both capital and time. Average Revenue from online orders were around Rs.150.
- ➢ Offline orders on the other hand, had to be offered at a cheaper rate at roughly Rs.100 as we needed the psychological impulse of the buyer on the purchase.
- ➢ In offline sales, we had to predict what the customers wanted for which we can never be accurate. This resulted in some posters remaining unsold which had to be sold at throwaway prices the next day.
- ➢ Both methods have their own merits, which is why we attempted to combine both. On the first day of our stalls, we accepted custom orders to be delivered the next day. This enabled us to offer the customer greater options with the reliability we can offer in offline sales.

❖ **Promotion and Marketing techniques**

- ➢ In order to establish our brand and create brand recognition, we adopted multiple marketing techniques.
- ➢ The most relevant being giveaways. We launched a market survey where 4 lucky respondents would get a free poster. We randomly selected the winners, who were contacted and free posters were delivered to them. This helped create a word of mouth about our company to their peers.
- ➢ We, on a trial basis for a very short term, also offered freebies like chocolates to attract more customers but we soon realised that it was not very effective.

Limitations of the Business

It is safe for one to say that Bandar Mifflin Poster Company was a successful business venture. Each stakeholder invested ₹250 and ended up with ₹2960, a 1084% return on initial investment. Bandar Mifflin also broke even after selling 3 units.

That being said, no business is perfect, and there remain flaws that need to be addressed.
The first limitation arises in the networking aspect of the business. If our team wasn't limited to four individuals, we would've had more contacts, made more connections, understood the market better, and identified more potential customers. An increase in social networking translates to more public interest in our company, more revenue, and thus, more profit.

The second limitation was our misjudgement of time. We sold the most units during our allotted time running the stall. We would've sold even more had our time at the stall not been limited to two days. Additionally, we generated designs and posted them too late, posting them in the first week of January, missing out on hypothetical online orders and sales before January. Not to mention that we would've increased the diversity in our poster designs and consumer choice had we started posting online and taking orders from December.

Limitations Of Analysis

The first limitation in our analysis lies in the fact that we were operating with very little market data. The market survey we conducted was insufficient, as we asked questions that were too broad in nature, leading to a lack of understanding of what the market demands in terms of poster designs. This may have affected the depth and breadth of our analysis as we were operating with limited data.

Businesses are dynamic entities, and market conditions may change rapidly without warning. The analysis that we conducted on our business may not be reliable in the future as it captures a specific point in time. Essentially, the ethos of ceteris paribus does not apply. As the sands of time flow, all conditions will not remain constant.

Key Performance Indicators

Investment : Initial investment of Rs.1,000, Rs.250 contributed by each partner.
Total sales : Sold 146 units out of which 80 units offline
Total revenue : Rs. 16,070.
Total cost : Rs. 5,229
Operating Profit Margin (OPM) : 67.46%
Dividend : Each partner earned a dividend of Rs. 2,960.25

Conclusion

As per the words of Robert J Oppenheimer, “Theory will only take you so far” what he meant by this is that what the graphs or theories suggest on paper can’t always predict what will happen in real life. This is what we were able to learn in this project. Many of the graphs that we were able to procure do not match the theoretical shapes that we had been taught. We had a very enriching experience and were able to witness firsthand the theory of demand in play during the days of the stall, where we had to undercut prices to sell more while also competing with a rival firm. We experienced issues with our supplier, but in the end, we were able to bring smiles to the faces of our customers and investors who now could bring some life into their otherwise empty walls, and some money back into their otherwise empty bank accounts. What better to look at than what you like the most?

Satisfaction Guaranteed, And Definitely.

Acknowledgment

We would like to take the time to thank our professor, Dr. Aditi Sawant for giving us the opportunity to understand the world of business and commerce. All the information we were able to receive, and all the experience we had were truly enriching and valuable. We would also like to thank the owners of Super Digital Printing and Rohan Copy Centre for assisting us in this journey.



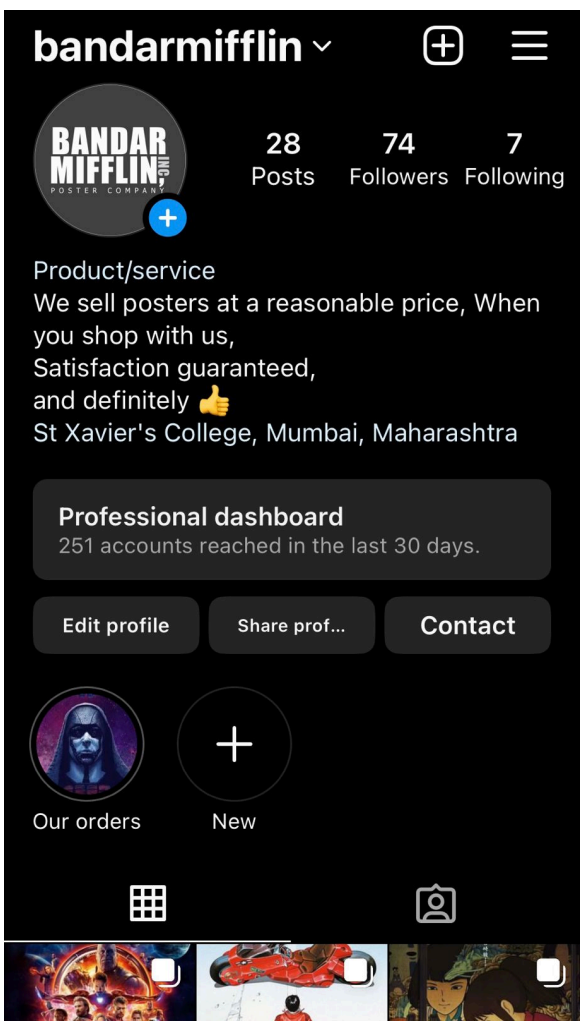
Appendix

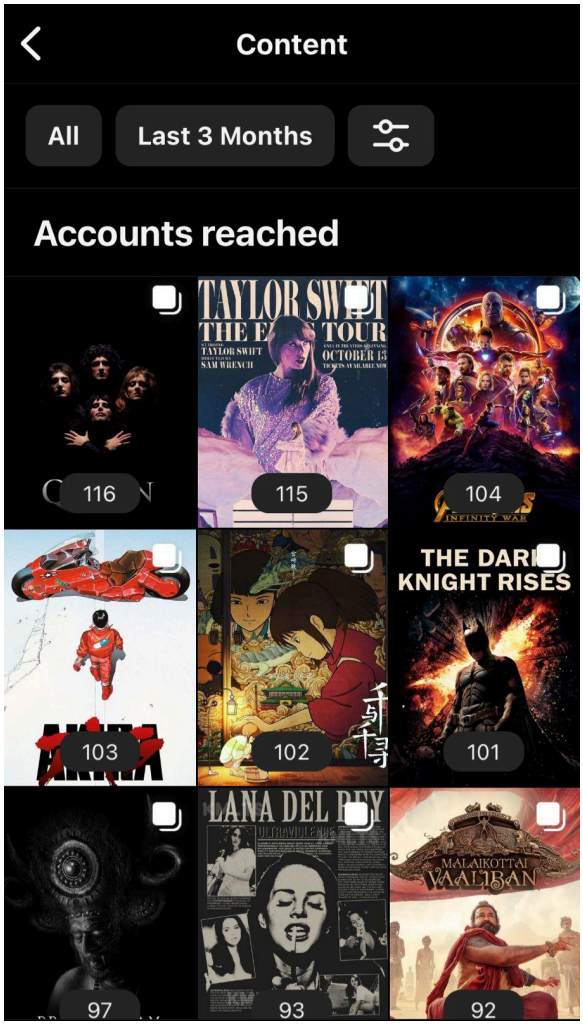
Appendix A - Cost and Revenue Sheet

	Units	TFC	TVC	TC	AC	MC
	0	₹366.00	₹0.00	₹366.00	-	-
	1	₹366.00	₹27.85	₹393.85	₹393.85	₹27.85
	2	₹366.00	₹55.70	₹421.70	₹210.85	₹27.85
	3	₹366.00	₹83.55	₹449.55	₹149.85	₹27.85
	4	₹366.00	₹111.40	₹477.40	₹119.35	₹27.85
	5	₹366.00	₹139.25	₹505.25	₹101.05	₹27.85
	6	₹366.00	₹167.10	₹533.10	₹88.85	₹27.85
	7	₹366.00	₹194.95	₹560.95	₹80.14	₹27.85
	8	₹366.00	₹218.28	₹584.28	₹73.04	₹23.33
Online sales	9	₹366.00	₹241.61	₹607.61	₹67.51	₹23.33
	10	₹366.00	₹264.94	₹630.94	₹63.09	₹23.33
	20	₹366.00	₹486.45	₹852.45	₹42.62	₹22.15
	30	₹366.00	₹709.23	₹1,075.23	₹35.84	₹22.28
Stall Day 1	33	₹366.00	₹774.16	₹1,140.16	₹34.55	₹21.64
	40	₹366.00	₹927.16	₹1,293.16	₹32.33	₹21.86
	50	₹366.00	₹1,147.31	₹1,513.31	₹30.27	₹22.02
	60	₹366.00	₹1,361.80	₹1,727.80	₹28.80	₹21.45
	70	₹366.00	₹1,582.10	₹1,948.10	₹27.83	₹22.03
	80	₹366.00	₹1,802.41	₹2,168.41	₹27.11	₹22.03
Stall Day 2	89	₹366.00	₹2,002.90	₹2,368.90	₹26.62	₹22.28
	90	₹366.00	₹2,025.60	₹2,391.60	₹26.57	₹22.70
	100	₹366.00	₹2,249.20	₹2,615.20	₹26.15	₹22.36
Post stall sale	106	₹366.00	₹2,378.60	₹2,744.60	₹25.89	₹21.57
	110	₹366.00	₹2,654.09	₹3,020.09	₹27.46	₹68.87
	120	₹366.00	₹3,247.28	₹3,613.28	₹30.11	₹59.32
	130	₹366.00	₹3,868.71	₹4,234.71	₹32.57	₹62.14
	140	₹366.00	₹4,493.17	₹4,859.17	₹34.71	₹62.45
Bangalore	146	₹366.00	₹4,863.00	₹5,229.00	₹35.82	₹61.64

Units	TR	AR	MR	Profit	Marginal Profit
0	₹0.00	-	-	-₹366.00	-
1	₹130.00	₹130.00	₹130.00	-₹263.85	₹102.15
2	₹260.00	₹130.00	₹130.00	-₹161.70	₹102.15
3	₹480.00	₹160.00	₹220.00	₹30.45	₹192.15
4	₹590.00	₹147.50	₹110.00	₹112.60	₹82.15
5	₹740.00	₹148.00	₹150.00	₹234.75	₹122.15
6	₹890.00	₹148.33	₹150.00	₹356.90	₹122.15
7	₹1,020.00	₹145.71	₹130.00	₹459.05	₹102.15
8	₹1,150.00	₹143.75	₹130.00	₹565.72	₹106.67
9	₹1,300.00	₹144.44	₹150.00	₹692.39	₹126.67
10	₹1,400.00	₹140.00	₹100.00	₹769.06	₹76.67
20	₹2,297.43	₹114.87	₹89.74	₹1,444.98	₹67.59
30	₹3,174.53	₹105.82	₹87.71	₹2,099.30	₹65.43
33	₹3,440.00	₹104.24	₹88.49	₹2,299.84	₹66.85
40	₹4,020.15	₹100.50	₹82.88	₹2,726.99	₹61.02
50	₹4,833.93	₹96.68	₹81.38	₹3,320.62	₹59.36
60	₹5,640.70	₹94.01	₹80.68	₹3,912.90	₹59.23
70	₹6,437.50	₹91.96	₹79.68	₹4,489.40	₹57.65
80	₹7,264.31	₹90.80	₹82.68	₹5,095.90	₹60.65
89	₹7,970.00	₹89.55	₹78.41	₹5,601.10	₹56.13
90	₹8,073.67	₹89.71	₹103.67	₹5,682.07	₹80.97
100	₹9,068.97	₹90.69	₹99.53	₹6,453.77	₹77.17
106	₹9,670.00	₹91.23	₹100.17	₹6,925.40	₹78.61
110	₹10,330.02	₹93.91	₹165.01	₹7,309.93	₹96.13
120	₹11,911.58	₹99.26	₹158.16	₹8,298.30	₹98.84
130	₹13,523.00	₹104.02	₹161.14	₹9,288.29	₹99.00
140	₹15,133.70	₹108.10	₹161.07	₹10,274.53	₹98.62
146	₹16,070.00	₹110.07	₹156.05	₹10,841.00	₹94.41

Appendix B - Photo Of Instagram page





Appendix C - Photos of business

Photo of team members



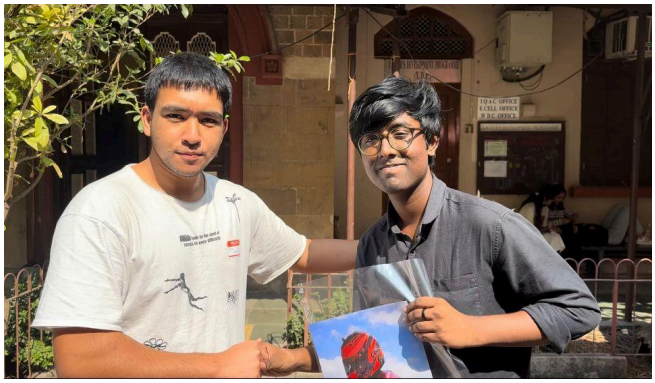
(left to right) Sidharth S, Jonathan Marattukalam, Chris Kurian Joseph, Basil Shibu



Inducing customers



Employee satisfaction

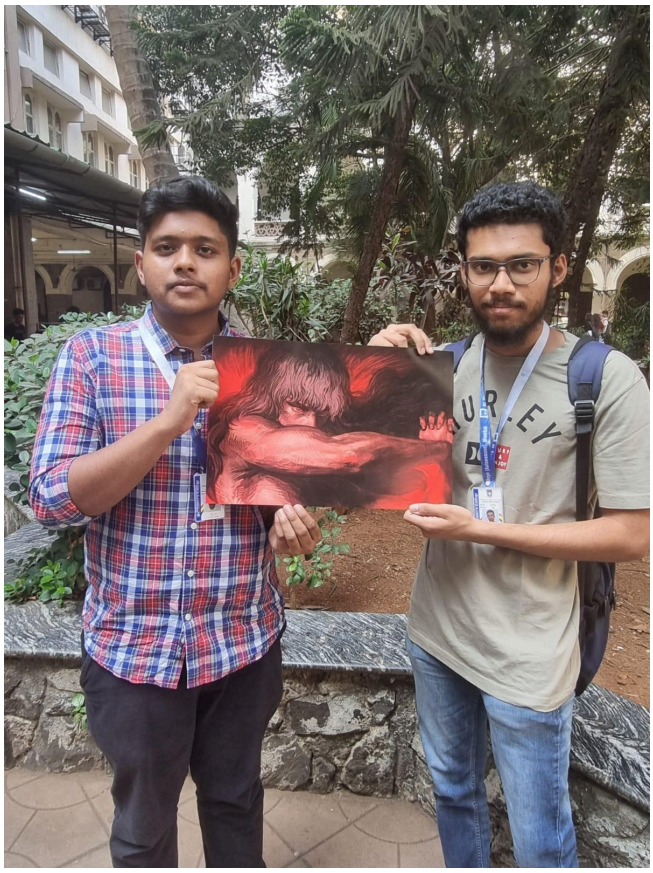


Our last in-person sale



Photo Of Our Stall

Our First Sale



Appendix D - Market Survey Questionnaire

POSTER COMPANY SURVEY

This survey is a part of our economics project for this term to start our own business, the four of us have decided to start our own poster company where we sell posters of various movies, shows, music album covers, anime and even posters of sportspersons and teams. This survey is to see what kind of posters you'd be interested in buying.

Email *

Valid email

This form is collecting emails. Change settings

Three lucky participants will receive a free poster, please enter your WhatsApp number below:

Short answer text

What form of entertainment do you enjoy? *

- [ ] Movies
- [ ] TV Shows/ Sitcoms
- [ ] Music
- [ ] Sports
- [ ] Anime
- [ ] NONE
- [ ] K Drama
- [ ] Books

Which of the following movie industries do you watch movies from? *

- [ ] Hindi Film industry
- [ ] Hollywood
- [ ] Telugu Film industry
- [ ] Tamil Film Industry
- [ ] Kannada Film Industry
- [ ] Malayalam Film Industry
- [ ] Korean Film Industry
- [ ] NONE
- [ ] Marathi Film Industry

Which of the following OTT platform do you use the most? *

- ( ) Netflix - Stranger things, FRIENDS, Drive to Survive, Cobra Kai, Breaking Bad etc.
- ( ) Amazon Prime - The Boys, The rings of power, Gen V, Jack Ryan, Farzi etc.
- ( ) Disney+ Hotstar - All MCU content, Modern Family, Family Guy, The Simpsons etc.
- ( ) NONE
- ( ) Other...

Which of the following film franchises do you love the most? *

- ( ) MCU
- ( ) DCU
- ( ) Harry Potter
- ( ) Star Wars
- ( ) Lord of the Rings
- ( ) LCU
- ( ) YRF Spy Universe
- ( ) Rohit Shetty Cop Universe
- ( ) NONE
- ( ) Other...

What video game titles are you interested in?

- [ ] Grand Theft Auto
- [ ] Call Of Duty
- [ ] FIFA
- [ ] Fortnite
- [ ] God Of War
- [ ] Spider-Man
- [ ] Last Of Us
- [ ] Red-Dead Redemption

What genres of music do you listen to *

- [ ] POP
- [ ] ROCK
- [ ] JAZZ
- [ ] BOLLYWOOD
- [ ] REGIONAL
- [ ] RAP
- [ ] R AND B
- [ ] COUNTRY
- [ ] HIP HOP
- [ ] CLASSICAL
- [ ] K-POP
- [ ] None

Which artists do you listen to? *

- [ ] Taylor Swift
- [ ] Kanye West
- [ ] Pritam
- [ ] Anirudh Ravichander
- [ ] Shreya Goshal
- [ ] Arjit Singh
- [ ] Ariana Grande
- [ ] Lana Del Rey
- [ ] Billie Eilish
- [ ] BTS
- [ ] Travis Scott
- [ ] Tyler, the creator
- [ ] Other...

Which sports do you watch? *

- ( ) Cricket
- ( ) Football
- ( ) Basketball
- ( ) Badminton
- ( ) Tennis
- ( ) F1
- ( ) None
- ( ) MMA
- ( ) Boxing
- ( ) Other...

Who is your favorite athlete?

Short answer text

What Anime do you watch?

- [ ] Naruto
- [ ] One Piece
- [ ] Jujutsu Kaisen
- [ ] Attack on Titan
- [ ] Death Note
- [ ] Berserk
- [ ] Other...

At what price would you be willing to buy a poster, keep in mind that a higher price range ensures a bigger size such as A3 and better quality thick paper, while the lower ranges will be A4 and on thin paper. *

- ( ) 180-200
- ( ) 150-180
- ( ) 120-150

Is there any specific poster that you would definitely buy at any of the prices given above if we were to print them for you?

Short answer text

Appendix E - Profit and Loss statement

Bandar Mifflin Poster Company	
Consolidated Profit and Loss statement	
Dec 2023 - Jan 2024	
Sales	
Sale of posters	₹16,070.00
Operating Costs	
Cost of Goods Sold	₹3,263.00
Commission payment	₹1,600.00
Research and Development	₹55.00
General and Administrative	₹181.00
Marketing	₹130.00
Total Operating Cost	₹5,229.00
Operating Profit	₹10,841.00
Net Profit	₹10,841.00
OPM	67.46%
NPM	67.46%